

Convergence of a numerical scheme for solving an inverse problem : Application to tsunami reconstruction

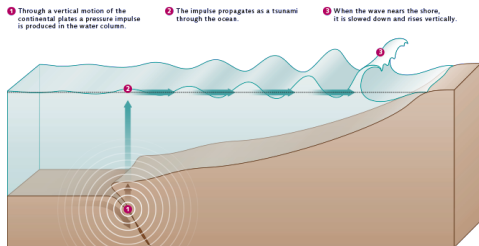
PICOF'2025

Raphaël Terrine (ENSTA-CMAP)

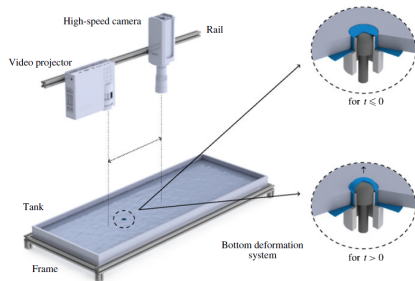
Collaborators : Laurent Bourgeois (UMA) and Philippe Moireau (CMAP)



Motivation



(a) Formation of a tsunami

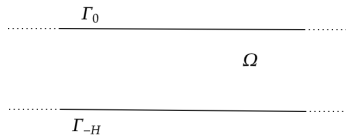


(b) Experimental setup

- Tsunami : seismic wave generated by the displacement of the sea bottom.
- Challenge : quickly detect the sea bottom displacement before disaster.
- Already studied through Tikhonov mixed formulation in the harmonic case, L. Bourgeois, J-F. Mercier, R. Terrine, 2024.
- Problem studied experimentally in a tank, Pablo J. Cobelli et al, 2017.

- 1. Models and studied problem**
- 2. Functional minimisation in an abstract framework**
- 3. Numerical results**
- 4. Conclusions and perspectives**

Complete 2D model and asymptotic limit



g : gravity

c : acoustic celerity

Complete 2D model and asymptotic limit

$$\begin{array}{c} \text{.....} \Gamma_0 \text{.....} \\ \Omega \\ \text{.....} \Gamma_{-H} \text{.....} \end{array} \quad \begin{array}{l} g : \text{gravity} \\ c : \text{acoustic celerity} \end{array}$$

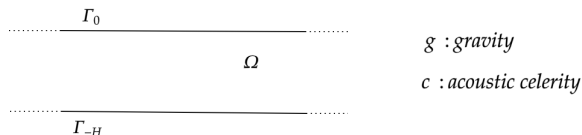
→ Obtained through linearisation of Euler equations^a:

$$\left\{ \begin{array}{ll} (\Delta - (1/c^2)\partial_t^2) \phi = 0 & \text{in } \Omega \times (0, T), \\ \partial_y \phi = \partial_t \textcolor{blue}{s} & \text{on } \Gamma_0 \times (0, T), \\ \partial_t \phi + g \textcolor{blue}{s} = 0 & \text{on } \Gamma_0 \times (0, T), \\ \partial_\nu \phi = -\partial_t \zeta := \textcolor{red}{\chi} & \text{on } \Gamma_{-H} \times (0, T), \\ \phi(0) = \partial_t \phi(0) = 0 & \text{in } \Omega. \end{array} \right.$$

- ϕ : velocity potential.
- $\textcolor{blue}{s}$: perturbation of ocean free surface.
- $\textcolor{red}{\chi}$: sea-bottom velocity.

^a J. Dubois et al, Well-posedness and potential-based formulation for the propagation of waves, 2024

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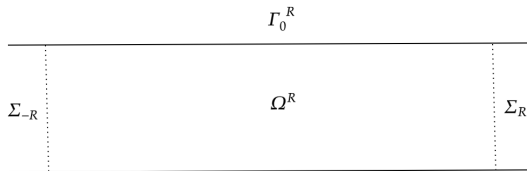
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Two models :

- Complete model $\implies c < \infty$.
- Gravity model $\implies c = \infty$.

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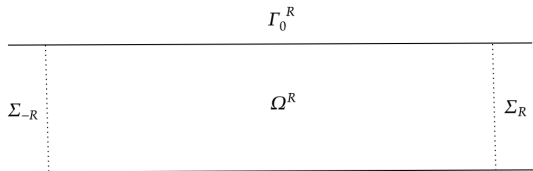
Bounded domain



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¹K. Dgaygui and P. Joly, *Absorbing boundary conditions for linear gravity waves*, 2006

Bounded domain



$$\left\{ \begin{array}{ll} \left(\Delta - (1/c^2) \partial_t^2 + \nu \Delta \partial_t \right) \phi = 0 & \text{in } \Omega^R, \\ \partial_y \phi + \nu \partial_\nu \partial_t \phi = \partial_t s & \text{on } \Gamma_0^R, \\ \partial_t \phi + g s = 0 & \text{on } \Gamma_0^R, \\ \partial_\nu \phi + \nu \partial_\nu \partial_t \phi = \chi & \text{on } \Gamma_{-H}^R, \\ \pm (\partial_x \phi + \nu \partial_\nu \partial_t \phi) + \frac{1}{c} \partial_t \phi = 0 & \text{on } \Sigma_{\pm R}^2, \\ \phi(0) = \partial_t \phi(0) = 0 & \text{in } \Omega^R. \end{array} \right.$$

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Let V, H, U three Hilbert spaces such that $V \hookrightarrow H \equiv H' \hookrightarrow V'$.

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Assumptions :

- For $z, \tilde{z} \in V$, $a_\nu(z, \tilde{z}) = \langle A_\nu z, \tilde{z} \rangle_{V', V}$ and a_ν continuous and λ -coercive.
- Source term $\chi \in L^2(0, T; U)$.
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Theorem

The problem has a unique solution z^a and:

$$\|z\|_{W_0(0, T)} \leq c \|\chi\|_{L^2(0, T; U)}.$$

^a J-L. Lions E. Magenes, *Non homogeneous boundary value problems and applications*, 1972

Functional framework: order 1 system

- $H = H^1(\Omega^R) \times L^2(\Omega^R) \times L^2(\Gamma_0^R)$ with the following norm

$$\|(\phi, \psi, s)^T\|_H^2 = \|\phi\|_{H^1(\Omega^R)}^2 + \frac{1}{c^2} \|\psi\|_{L^2(\Omega^R)}^2 + g \|s\|_{L^2(\Gamma_0^R)}^2,$$

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Abstract form

The regularised problem solution $z = (\phi, \psi, s)^T \in W_0(0, T)$ is a solution in the variational sense of :

$$\dot{z} + A_\nu z = B\chi \text{ in } L^2(0, T; V').$$

Optimal control approach for solving the inverse problem

Objective

Recovering χ through the measurement y of s in $L^2(0, T, L^2(\Gamma_0^R))$.

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Tikhonov regularisation : For $\epsilon > 0$:

$$\inf_{\chi \in L^2(0, T; L^2(\Gamma_{-H}^R))} \mathcal{J}(\chi) = \int_0^T \frac{1}{2} \|y(t) - Cz|_{\chi}(t)\|_{L^2(\Gamma_0^R)}^2 + \frac{\epsilon}{2} \|\chi(t)\|_{L^2(\Gamma_{-H}^R)}^2 dt,$$

with $\forall z = (\phi, \psi, s)^T \in H$, $Cz = s$ and

$$\begin{cases} \dot{z}|_{\chi} + A_{\nu} z|_{\chi} = B\chi, \\ z|_{\chi}(0) = 0, \end{cases}$$

Optimal control approach for solving the inverse problem : abstract framework

Let Y, V, H, U four Hilbert spaces with $V \subset H \subset V'$. Let $C \in \mathcal{L}(H, Y)$: measurement operator.

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Recover χ through the measurement y of Cz .

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Well-posedness

The previous problem has a unique solution : $\mathcal{J}$ continuous, coercive, strictly convex.

Recovering χ

Lagrangian :

$$\min_{\substack{z \in W_0(0,T) \\ \chi \in L^2(0,T;U)}} \max_{p \in L^2(0,T;V)} \mathcal{L}(z, p, \chi) = \int_0^T \frac{1}{2} \|y(t) - Cz(t)\|_Y^2 + \frac{\epsilon}{2} \|\chi(t)\|_U^2 + \langle \dot{z} + A_\nu z - B\chi, p \rangle_{V',V},$$

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Saddle points of continuous problem

The saddle points $(\bar{z}, \bar{p}, \bar{\chi})$ satisfy:

$$\begin{cases} \dot{\bar{z}} + A_\nu \bar{z} = B\bar{\chi} \text{ in } L^2(0, T; V'), \\ \dot{\bar{p}} - A_\nu^* \bar{p} = C^*(C\bar{z} - y) \text{ in } L^2(0, T; V'), \\ \bar{z}(0) = \bar{p}(T) = 0 \text{ in } H, \\ \bar{\chi} = \frac{1}{\epsilon} B^* \bar{p}. \end{cases} \quad (P_c)$$

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"Optimise then discretise" or "Discretise then optimise" ?

Adjoint state of discretised criterion

$$\mathcal{J}_\tau^h(\{\chi_h^n\}_{1 \leq n \leq N}) = \sum_{n=0}^N \frac{\tau}{2} \|y_h^n - Cz_h^n\|_Y^2 + \sum_{n=1}^N \frac{\tau \epsilon}{2} \|\chi_h^n\|_U^2,$$

$$\text{with } \begin{cases} \frac{z_h^n - z_h^{n-1}}{\tau} + A_\nu z_h^n - B\chi_h^n = 0, & \forall n \in \llbracket 1; N \rrbracket \\ z_h^0 = 0. \end{cases}$$

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Solving the problem as an engineer

- Compute analytically the gradient of $\mathcal{J}_\tau^h$.
- Use a gradient descent algorithm to find the minimiser.
- We use L-BFGS for fast convergence.

Parameters

- $R = 90cm$.
- $H = 15cm$.
- $h = 0.1, \mathbb{P}_2$ elements.
- $\tau = h^2$.
- $\epsilon = \nu = 10^{-6}$.

Our contribution : Numerical analysis of the scheme

$$\begin{cases} \dot{\bar{z}} + A_\nu \bar{z} = B\bar{\chi} \text{ in } L^2(0, T; V'), \\ \dot{\bar{p}} - A_\nu^* \bar{p} = C^*(C\bar{z} - y) \text{ in } L^2(0, T; V'), \\ \bar{z}(0) = \bar{p}(T) = 0 \text{ in } H, \\ \bar{\chi} = \frac{1}{\epsilon} B^* \bar{p}. \end{cases}$$

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Questions

- $(\{z_h^n\}_{0 \leq n \leq N}, \{p_h^n\}_{1 \leq n \leq N}, \{\chi_h^n\}_{1 \leq n \leq N}) \rightarrow (\bar{z}, \bar{p}, \bar{\chi})$ when $(\tau, h) \rightarrow 0$?
- At which rate ?

Space-time mixed formulation

Find $u = (z, \chi)^T \in W_0(0, T) \times L^2(0, T, U)$ and $p \in L^2(0, T; V)$ such that for all $v = (\tilde{z}, \eta)^T \in W_0(0, T) \times L^2(0, T, U)$ and for all $q \in L^2(0, T; V)$:

$$\begin{cases} j(u, v) + d(v, p) = \ell(v), \\ d(u, q) = 0, \end{cases}$$

$$j(u, v) = \int_0^T \epsilon(\chi, \eta)_U + (Cz, C\tilde{z})_Y \, dt, \quad d(u, q) = \int_0^T \langle \dot{z}, q \rangle_{V', V} + a_\nu(z, q) - b(\chi, q) \, dt$$
$$\ell(v) = \int_0^T (y, C\tilde{z})_Y \, dt.$$

Space-time mixed formulation

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$$\begin{cases} j(u, v) + d(v, p) = \ell(v), \\ d(u, q) = 0, \end{cases}$$

$$j(u, v) = \int_0^T \epsilon(\chi, \eta)_U + (Cz, C\tilde{z})_Y \, dt, \quad d(u, q) = \int_0^T \langle \dot{z}, q \rangle_{V', V} + a_\nu(z, q) - b(\chi, q) \, dt$$
$$\ell(v) = \int_0^T (y, C\tilde{z})_Y \, dt.$$

Equivalence between problems

(z, p, χ) solutions of the mixed formulation are the solutions $(\bar{z}, \bar{p}, \bar{\chi})$ of (P_c) .

Well-posedness of the mixed formulation

Coercivity of j on $\ker d$

There exists $C > 0$ such that for all $u \in W_0(0, T) \times L^2(0, T; U)$ such that $d(u, q) = 0, \forall q \in L^2(0, T; V)$, we get :

$$j(u, u) \geq C \|u\|_{W_0(0, T) \times L^2(0, T; U)}^2$$

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Inf-sup condition

There exists $\beta > 0$ such that:

$$\inf_{p \in L^2(0, T; V)} \sup_{v \in W_0(0, T) \times L^2(0, T; U)} \frac{d(v, p)}{\|v\|_{W_0(0, T) \times L^2(0, T; U)} \|p\|_{L^2(0, T; V)}} \geq \beta.$$

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Well-posedness

The mixed formulation is well-posed ^a.

^aA.Ern J-L.Guermond, *Theory and practice of finite elements*, 2004

Finite elements discretisation

- **Discretisation spaces** : $V_h \subset V$, $U_h \subset U$ et $Y_h \subset Y$ for $h > 0$.
- $W_h^\tau = \{w_h^\tau \in \mathbb{P}_1 \otimes V_h, w_h^\tau(0) = 0\}$.

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Discrete mixed formulation

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$$z_h^\tau(x, t) = \sum_{n=1}^N z_h^n(x) \omega^n(t); \quad \chi_h^\tau(x, t) = \sum_{n=1}^N \chi_h^n(x) \omega_0^n(t); \quad y_h^\tau(x, t) = \sum_{n=1}^N y_h^n(x) \omega_0^n(t), \quad p_h^\tau(x, t) = \sum_{n=1}^N p_h^n(x) \omega_0^n(t).$$

Conforming time scheme

Time scheme with conforming FEM

$$\begin{cases} \frac{z_h^n - z_h^{n-1}}{\tau} + A_\nu \frac{z_h^n + z_h^{n-1}}{2} = B\chi_h^n, & \forall n \in \llbracket 1; N \rrbracket, \\ \frac{p_h^n - p_h^{n+1}}{\tau} + A_\nu^* \left(\frac{p_h^{n+1} + p_h^n}{2} \right) = C^* \left(\frac{y_h^{n-1} + y_h^n}{2} - C \frac{z_h^{n-1} + 4z_h^n + z_h^{n+1}}{6} \right), & \forall n \in \llbracket 1; N-1 \rrbracket \\ \frac{p_h^N - p_h^{N+1}}{\tau} + A_\nu^* \left(\frac{p_h^{N+1} + p_h^N}{2} \right) = C^* \left(\frac{y_h^{N-1} + y_h^N}{2} - C \left(\frac{1}{6} z_h^{N-1} + \frac{1}{3} z_h^N \right) \right), \\ \epsilon \chi_h^n - B^* p_h^n = 0, & \forall n \in \llbracket 1; N \rrbracket, \\ p_h^{N+1} = z_h^0 = 0. \end{cases}$$

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- Data attachment term different from that of the Euler implicit scheme.
- Crank-Nicholson scheme rather than implicit Euler scheme (no uniform energy identity).

Adding a quadrature rule

With $\oint_0^T f(t) dt = \tau \sum_{n=1}^N f(n\tau_-)$:

- $j(u_h^\tau, v_h^\tau) \rightarrow j_\tau(u_h^\tau, v_h^\tau) = \int_0^T \epsilon(\partial_t \chi_h^\tau, \partial_t \eta_h^\tau)_U dt + \oint_0^T (Cz_h^\tau, C\tilde{z}_h^\tau)_Y dt.$
- $\ell(v_h^\tau) \rightarrow \ell_\tau(v_h^\tau) = \oint_0^T (y, C\tilde{z}_h^\tau)_Y dt.$
- $d(u_h^\tau, q_h^\tau) \rightarrow d_\tau(u_h^\tau, q_h^\tau) = \int_0^T \langle \dot{z}_h^\tau, q_h^\tau \rangle_{V', V} - b(\chi_h^\tau, q_h^\tau) dt + \oint_0^T +a_\nu(z_h^\tau, q_h^\tau) dt.$

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$$\begin{cases} \frac{z_h^n - z_h^{n-1}}{\tau} + A_\nu z_h^n = B\chi_h^n, & \forall n \in \llbracket 1; N \rrbracket, \\ \frac{p_h^n - p_h^{n+1}}{\tau} + A_\nu^* p_h^n = C^*(y_h^n - Cz_h^n), & \forall n \in \llbracket 1; N \rrbracket \\ \epsilon \chi_h^n - B^* p_h^n = 0 & \forall n \in \llbracket 1; N \rrbracket, \\ p_h^{N+1} = z_h^0 = 0. \end{cases}$$

Technical results

Assumption : quasi-uniform mesh

We suppose that the mesh is "shape regular" and there exists $c > 0$ such that:

$$\forall h > 0, \forall K \in \mathcal{T}_h, ch \leq h_K := \text{Diam}(K) \leq h.$$

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For all $v_h \in V_h$, we introduce the following norm:

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Equivalence of norms

For all $v_h \in V_h$, we have $\|v_h\|_{V'_h} \leq \|v_h\|_{V'} \leq \|P_h\|_{\mathcal{L}(V)} \|v_h\|_{V'_h}$ with

$$P_h : V \rightarrow V_h, P_h(z) = \arg \min_{z_h \in V_h} \|z - z_h\|_H.^a$$

^aA.Ern J-L.Guermond, *Finite Elements III*, 2021

Well-posedness of the discrete problem

Discrete mixed formulation with quadrature

Find $u_h^\tau = (z_h^\tau, \chi_h^\tau)^T \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)$ and $p_h^\tau \in \mathbb{P}_0 \otimes V_h$ such that for all $v_h^\tau = (\tilde{z}_h^\tau, \eta_h^\tau)^T \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)$ and for all $q_h^\tau \in \mathbb{P}_0 \otimes V_h$:

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Well-posedness of the discrete problem

Discrete mixed formulation with quadrature

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- j_τ coercive on $\ker d_\tau$ with a constant α that depends on $\|P_h\|_{\mathcal{L}(V)}$.
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Well-posedness

The discrete mixed formulation with quadrature is well-posed.^a

^aF. Brezzi, M. Fortin, *Mixed and hybrid finite element methods*, 1991

Convergence rate

Second Strang Lemma

$$\begin{aligned} \|u_h^\tau - u\|_{W_0(0,T) \times L^2(0,T;U)} &\leq C \|P_h\|_{\mathcal{L}(V)}^3 \left(\inf_{v_h^\tau \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)} \|u - v_h^\tau\|_{W_0(0,T) \times L^2(0,T;U)} \right. \\ &+ \inf_{q_h^\tau \in \mathbb{P}_0 \otimes V_h} \|p - q_h^\tau\|_{L^\infty(0,T;V)} + \inf_{v_h^\tau \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)} \|u - v_h^\tau\|_{L^\infty(0,T;V) \times L^2(0,T;U)} \\ &+ \sup_{v_h^\tau \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)} \frac{|j(u, v_h^\tau) - j_\tau(u, v_h^\tau)|}{\|v_h^\tau\|_{W_0(0,T) \times L^2(0,T;U)}} + \sup_{v_h^\tau \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)} \frac{|d(v_h^\tau, p) - d_\tau(v_h^\tau, p)|}{\|v_h^\tau\|_{W_0(0,T) \times L^2(0,T;U)}} \\ &+ \left. \sup_{v_h^\tau \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)} \frac{|\ell(v_h^\tau) - \ell_\tau(v_h^\tau)|}{\|v_h^\tau\|_{W_0(0,T) \times H_1^1(0,T;U)}} + \sup_{q_h^\tau \in \mathbb{P}_0 \otimes V_h} \frac{|d(u, q_h^\tau) - d_\tau(u, q_h^\tau)|}{\|q_h^\tau\|_{L^2(0,T;V)}} \right) \end{aligned}$$

Convergence rate

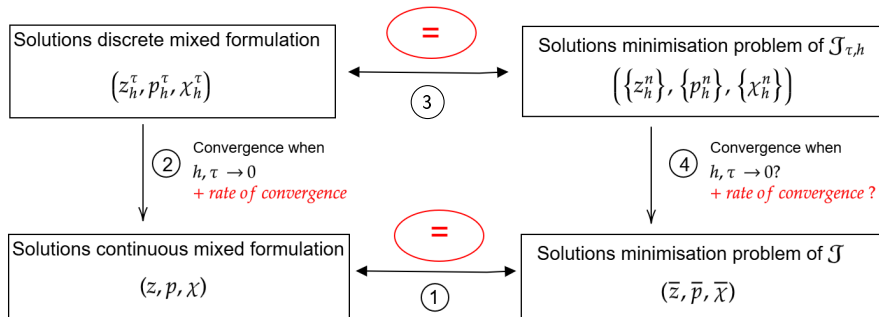
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Convergence error for $\mathbb{P}_k$ Lagrange finite elements

$$\sqrt{\|z_h^\tau - z\|_{W_0(0,T)}^2 + \|\chi_h^\tau - \chi\|_{L^2(0,T;U)}^2} \leq C \|P_h\|_{\mathcal{L}(V)}^3 (\tau + h^k)$$

Summary of the presentation



Summary

We seek to prove 4.

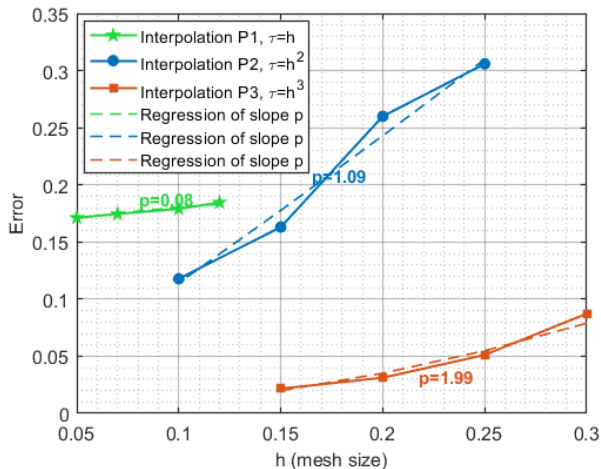
- We first prove 1.
- By adding a quadrature rule and by choosing the right discretisation, we prove 2.
- We then observe that 3 is verified.

Observed rate of convergence

$$\|\chi_h^\tau - \chi\|_{L^2(0,\tau;U)} \leq \frac{C}{h^{\frac{3}{2}}}(\tau + h^k)$$

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$$\|\chi_h^\tau - \chi\|_{L^2(0,T;U)} \leq \frac{C}{h^{\frac{3}{2}}}(\tau + h^k)$$



Future work

- $\nu = 0$.
- **Already tried** : Second order variational system -> Lack of inf-sup condition.

Other work

- Tikhonov/Morozov mixed formulation accelerated by SOR method.
- Links between the two methods.

Choice of ϵ with respect to the noise level

